BASE CHANGE FOR THE DERIVED PUSHFORWARD IN A SPECIAL CONTINUOUS AND COCONTINUOUS SQUARE OF RINGED TOPOI

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the reduction to abelian sheaves via condition (5), the bounded below injective-resolution argument with the acyclicity check and Leray's acyclicity lemma, and the unbounded K-injective argument relying on the exactness of g'_! in assumption (6).

Editorial note. The label lemma-special-square-continuous names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. Pre-open gate: python operations/agent-authoring/tools/precard_gate.py –source sites-cohomology.tex –label lemma-special-square-continuous –start 9594 –end 9681 –local-id 3555 returned RESULT=OK (L3555 free, occupancy FREE, public interval clear). Candidate taken from lane-s4/halfB-priority.json.

Editorial note. The statement contains a footnote recording when condition (6) holds; the footnote is delivered verbatim with the statement.

Editorial note. The statement and proof are copied from the pinned chapter file; the cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- Pullback along a cocontinuous morphism with trivial pullback of the structure sheaf. Same chapter, the lemmas identifying g^* with g^{-1} when the pullback of the structure sheaf is the structure sheaf. Supplies the identification g^* = g^{-1} and (g')^* = (g')^{-1} from condition (5), so the derived functors reduce to underived pullbacks.
- Vanishing of the higher cohomology of the pullback of an injective sheaf. Same chapter, the lemma that H^p(U', (g')^{-1}I^q) = 0 for p > 0 for an injective abelian sheaf I and a cocontinuous v'. Supplies the right acyclicity of the pullback of the terms of an injective resolution for the functor f'_*.
- Higher direct images as the sheaf associated to a presheaf of cohomology groups. Same chapter, the lemma identifying R^p f'_* with the sheafification of V' maps to H^p(u'(V'), -). Supplies the translation of the vanishing of the cohomology groups over the objects U' into the acyclicity statement used by Leray's acyclicity lemma.
- Leray's acyclicity lemma. Derived Categories, the lemma that a bounded below complex of objects acyclic for a left exact functor computes its right

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derived functor. Supplies the conclusion that f'_*(g')^*I^bullet represents Rf'_*(g')^{-1}K.
- The commutativity of the underlying pushforward and pullback in the special square. Modules on Sites, the lemma computing the pushforward of the pullback along the square involved for sheaves of modules. Supplies the identity f'_*(g')^{-1}I^bullet = g^{-1}f_*I^bullet, which identifies the two sides of the base change map.
- Adjoints of exact functors preserve K-injectivity. Derived Categories, the lemma that the right adjoint of an exact functor preserves K-injective complexes. Supplies, together with the exactness of g'_! in condition (6), the K-injectivity of (g')^{-1}I^bullet in the unbounded case.

Source reference. Original lemma 08BJ: sources/stacks-chow/tags.txt, source lines 10569–10569. Prerequisite pointer invoked at proof line(s) 9672; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 015E: sources/stacks-chow/tags.txt, source lines 1505–1505. Prerequisite pointer invoked at proof line(s) 9659; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 03Q8: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 747–747. Prerequisite pointer invoked at proof line(s) 9656; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0D5Y: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 4453–4453. Prerequisite pointer invoked at proof line(s) 9643; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0D6X: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 9278–9278. Prerequisite pointer invoked at proof line(s) 9651; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 04BH: sources/stacks-chow/tags.txt, source lines 5652–5652. Prerequisite pointer invoked at proof line(s) 9634; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0FN4: sources/stacks-chow/tags.txt, source lines 19554–19554. Prerequisite pointer invoked at proof line(s) 9663; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 00XN: sources/stacks-chow/tags.txt, source lines 1225–1225. Prerequisite pointer invoked at proof line(s) 9617; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 00X0: sources/stacks-chow/tags.txt, source lines 1202–1202. Prerequisite pointer invoked at proof line(s) 9616; it names a standard

fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Consider a commutative diagram*

$$\begin{array}{ccc} (Sh(\mathcal{C}'), \mathcal{O}_{\mathcal{C}'}) & \xrightarrow[(g', (g')^\sharp)]{} & (Sh(\mathcal{C}), \mathcal{O}_{\mathcal{C}}) \\ {\scriptstyle (f', (f')^\sharp)}\downarrow & & \downarrow{\scriptstyle (f, f^\sharp)} \\ (Sh(\mathcal{D}'), \mathcal{O}_{\mathcal{D}'}) & \xrightarrow{(g, g^\sharp)} & (Sh(\mathcal{D}), \mathcal{O}_{\mathcal{D}}) \end{array}$$

of ringed topoi and suppose we have functors

$$\begin{array}{ccc} \mathcal{C}' & \xrightarrow[v']{} & \mathcal{C} \\ {\scriptstyle u'}\uparrow & & \uparrow{\scriptstyle u} \\ \mathcal{D}' & \xrightarrow{v} & \mathcal{D} \end{array}$$

such that (with notation as in Sites, Sections 00X0 and 00XN) we have

(1) *u and u' are continuous and give rise to the morphisms f and f',*
(2) *v and v' are cocontinuous giving rise to the morphisms g and g',*
(3) *$u \circ v = v' \circ u'$,*
(4) *v and v' are continuous as well as cocontinuous, and*
(5) *$g^{-1}\mathcal{O}_{\mathcal{D}} = \mathcal{O}_{\mathcal{D}'}$ and $(g')^{-1}\mathcal{O}_{\mathcal{C}} = \mathcal{O}_{\mathcal{C}'}$.*

Then $Rf'_* \circ (g')^* = g^* \circ Rf_*$ as functors $D^+(\mathcal{O}_{\mathcal{C}}) \to D^+(\mathcal{O}_{\mathcal{D}'})$. If in addition

(6) *$g'_! : Ab(\mathcal{C}') \to Ab(\mathcal{C})$ is exact*[1],

then $Rf'_* \circ (g')^* = g^* \circ Rf_*$ as functors $D(\mathcal{O}_{\mathcal{C}}) \to D(\mathcal{O}_{\mathcal{D}'})$.

Proof. We have $g^* = Lg^* = g^{-1}$ and $(g')^* = L(g')^* = (g')^{-1}$ by condition (5). By Lemma 0D5Y it suffices to prove the result on the derived category $D^+(\mathcal{C})$ or $D(\mathcal{C})$ of abelian sheaves.

Choose an object $K \in D^+(\mathcal{C})$. Let $\mathcal{I}^\bullet$ be a bounded below complex of injective abelian sheaves on $\mathcal{C}$ representing K. By Lemma 0D6X we see that $H^p(U', (g')^{-1}\mathcal{I}^q) = 0$ for all $p > 0$ and any q and any $U' \in \mathrm{Ob}(\mathcal{C}')$. Recall that $R^p f'_*(g')^{-1}\mathcal{I}^q$ is the sheaf associated to the presheaf $V' \mapsto H^p(u'(V'), (g')^{-1}\mathcal{I}^q)$, see Lemma 03Q8. Thus we see that $(g')^{-1}\mathcal{I}^q$ is right acyclic for the functor f'_*. By Leray's acyclicity lemma (Derived Categories, Lemma 015E) we find that $f'_*(g')^*\mathcal{I}^\bullet$ represents $Rf'_*(g')^{-1}K$. By Modules on Sites, Lemma 0FN4 we find that $f'_*(g')^{-1}\mathcal{I}^\bullet = g^{-1}f_*\mathcal{I}^\bullet$. Since $g^{-1}f_*\mathcal{I}^\bullet$ represents $g^{-1}Rf_*K$ we conclude.

Choose an object $K \in D(\mathcal{C})$. Let $\mathcal{I}^\bullet$ be a K-injective complex of abelian sheaves on $\mathcal{C}$ representing K. By Derived Categories, Lemma 08BJ and assumption (6) we find that $(g')^{-1}\mathcal{I}^\bullet$ is a K-injective complex of abelian sheaves on $\mathcal{C}'$. By Modules on Sites, Lemma 0FN4 we find that $f'_*(g')^{-1}\mathcal{I}^\bullet = g^{-1}f_*\mathcal{I}^\bullet$. Since $f_*\mathcal{I}^\bullet$ represents Rf_*K and since $f'_*(g')^{-1}\mathcal{I}^\bullet$ represents $Rf'_*(g')^{-1}K$ we conclude. □

[1] Holds if fibre products and equalizers exist in $\mathcal{C}'$ and v' commutes with them, see Modules on Sites, Lemma 04BH.